Implicit Regularization in Nonlinear In-Context Learning

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Abstract

Recent theoretical advances have established that linear transformers perform in-context learning (ICL) by implementing gradient descent on linear regression tasks, often equivalent to ridge regression. However, real-world language models employ nonlinear activations (e.g., ReLU, GeLU) and softmax attention, whose implicit regularization effects remain poorly understood. This paper provides the first systematic theoretical analysis of implicit regularization in nonlinear ICL. We prove that a single-layer transformer with nonlinear MLP and softmax attention implicitly minimizes a kernel ridge regression objective in a reproducing kernel Hilbert space (RKHS) defined by the attention mechanism. We derive explicit generalization bounds that depend on the spectral properties of the attention kernel and the Lipschitz constant of the activation function. Furthermore, we identify a phase transition phenomenon: when the context length exceeds a critical threshold determined by the effective dimension of the kernel, the implicit regularization shifts from ℓ_2-norm penalty to a sparsity-inducing penalty akin to ℓ_1 regularization. Our theoretical findings are validated through synthetic experiments on polynomial regression and image classification tasks, demonstrating close alignment between empirical test error and our theoretical bounds.

Keywords: In-context learning, implicit regularization, generalization bounds, kernel methods, transformers, nonlinear activation

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

1.1. The Puzzle and Promise of In-Context Learning

In-context learning (ICL) stands as one of the most remarkable and somewhat mysterious capabilities of modern large language models (LLMs) (?). Unlike traditional fine-tuning, ICL allows a model to adapt to new tasks solely through inference-time conditioning on a few input-output examples provided in the prompt, without any parameter updates. This meta-learning-like behavior emerges unexpectedly during pre-training on next-token prediction and has become a cornerstone of practical LLM deployment.

The theoretical community has made significant strides in demystifying this phenomenon, particularly for linear transformer architectures. Seminal works by (?), (?), and others have elegantly shown that linear attention heads can implement gradient descent on a least-squares objective, making ICL equivalent to solving ridge regression. This line of work paints a compelling picture: ICL in linear models is essentially implicit Bayesian inference or gradient-based optimization performed in the forward pass.

However, reality is nonlinear. Every transformer block that powers today's LLMs contains two fundamentally nonlinear components: the softmax attention mechanism and the multi-layer perceptron (MLP) with nonlinear activations like ReLU or GeLU. While the linear theory is beautiful and instructive, it necessarily ignores these core architectural elements. We are thus left with a critical gap: if ICL in linear models is ridge regression, what is ICL in real, nonlinear transformers? What kind of learning algorithm does it implicitly execute, and what statistical guarantees can we derive?

Understanding the implicit regularization induced by these nonlinearities is not merely an academic exercise. It is the key bridge connecting the elegant but simplified linear theories to the messy, powerful reality of practical models. This understanding could illuminate why certain prompts work better than others, guide architecture design, and ultimately help us build more reliable and efficient in-context learners.

1.2. The Limits of Linearity and Our Position

Existing theoretical work provides essential pieces, but not the complete puzzle for nonlinear ICL.

Linear ICL Theory. The equivalence between linear transformers and gradient descent/ridge regression is a foundational result (???). These analyses, however, rely crucially on the assumption of linear attention or linearized approximations. They cannot capture the expressive power or inductive biases introduced by nonlinear activations. As noted in (?), the linear theory fundamentally cannot explain how models perform ICL on nonlinear tasks.

Classical Implicit Regularization. There is a rich literature on how gradient-based optimization imparts implicit biases, such as convergence to max-margin solutions in linear models (?) or low-rank solutions in matrix factorization (?). The Neural Tangent Kernel (NTK) theory (??) further shows that infinite-width networks trained with gradient descent behave like kernel regression. However, these analyses typically focus on a fixed dataset and a single task. ICL, in stark contrast, is a meta-learning scenario: the model is trained on a distribution of tasks, and at test time, the "dataset" (the in-context examples) is dynamic and task-specific. The object of regularization is not a single function but the learning algorithm itself.

Attention as Kernel Methods. The connection between softmax attention and kernel smoothing has been explored (??), and recent work by (?) has framed ICL as kernel gradient descent in a feature space defined by attention. Yet, these works often treat the attention kernel in isolation or within linearized settings. They do not provide a unified analysis of the full transformer block—where attention and MLP interact—and its consequent implicit regularization.

Emerging Explorations of Nonlinearity. Recent studies have begun to probe the necessity of nonlinearity. For instance, (?) empirically demonstrate that the MLP is crucial for nonlinear ICL, while linearized transformers fail. (?) provide initial steps by analyzing shallow ReLU networks in an ICL setting. However, a systematic theoretical framework that characterizes the implicit regularization of standard transformer architectures (softmax attention + nonlinear MLP) and derives finite-sample generalization bounds remains absent.

This paper aims to fill this exact gap. We move beyond the linear paradigm to establish the first comprehensive theory of implicit regularization in nonlinear in-context learning.

1.3. Our Contributions

We present a unified theoretical framework that characterizes how a single-layer transformer with softmax attention and a nonlinear MLP performs ICL. Our main contributions are fourfold:

1. A Kernel Representation Theorem for Nonlinear ICL. We prove that gradient flow training of a transformer induces a predictor that solves a kernel ridge regression (KRR) problem. Crucially, the kernel is not arbitrary; it is a structured combination $\alpha K_{\text{att}} + \beta K_{\text{mlp}}$, where K_{att} is an exponential kernel induced by the attention mechanism and K_{mlp} is the ReLU neural tangent kernel (Theorem 1). This provides a precise functional characterization of the nonlinear ICL algorithm.

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2. Tight Generalization Bounds. We derive non-asymptotic, high-probability upper bounds on the excess risk of the in-context learner (Theorem 2). These bounds explicitly depend on interpretable quantities:

   - The effective dimension d_{eff} of the combined kernel, capturing task complexity.
   - The Lipschitz constant of the activation function (e.g., 1 for ReLU).
   - The number of in-context examples n.

   The bound decays as $O(\sqrt{d_{\text{eff}}/n})$, providing a rigorous sample complexity guarantee for nonlinear ICL.

3. Discovery of a Regularization Phase Transition. We identify a surprising phase transition in the implicit regularization mechanism (Theorem 3). There exists a critical context length $n_c \propto d_{\text{eff}}$. When $n < n_c$ (under-specified regime), the implicit bias resembles an ℓ_2-norm penalty, promoting smooth solutions. When $n > n_c$ (over-specified regime), the bias morphs into an ℓ_1-type penalty, encouraging sparse representations in the kernel basis. This phenomenon explains how transformers can adapt their inductive bias based on available context.

4. Empirical Validation and Practical Insights. We validate all theoretical predictions on synthetic tasks (polynomial regression, RBF functions). The empirical risk curves closely match our derived bounds, and we clearly observe the predicted phase transition. Our theory yields actionable insights:

- Prompt Design: For simple tasks (low d_{eff}), short prompts suffice. For complex tasks, prompts must exceed the critical length n_c to activate the beneficial sparse regime.
- Architecture Guidance: The width of the MLP controls the richness of K_{mlp}, while the number of attention heads diversifies K_{att}.
- Meta-Training: The task distribution during pre-training shapes the effective kernel and thus the sample efficiency on downstream tasks.

In essence, this paper constructs a bridge from the well-understood land of linear ICL to the complex continent of nonlinear transformers. We show that the path across this bridge is illuminated by kernel methods and spectral analysis, revealing a landscape richer and more dynamic than previously imagined—one where the learning algorithm itself evolves with the length of the prompt.

1.4. Paper Organization

The rest of the paper is structured as follows. Section 2 reviews related work. Section 3 formalizes our problem setup and model architecture. Our main theoretical results—Theorems 1, 2, and 3—are presented in Section 4, with proof sketches. Detailed technical analyses and full proofs are deferred to Sections 5 and the Appendix. Section 6 presents empirical validations. We discuss broader implications, limitations, and future directions in Section 7, and conclude in Section 8.

2. Related Work

Our work sits at the intersection of several rapidly evolving research threads: the theoretical mechanics of in-context learning (ICL), implicit regularization in deep learning, and the kernel perspective on attention architectures. We position our contribution by tracing these lineages and highlighting the specific gap we address.

2.1. Theoretical Mechanisms of In-Context Learning

The quest to understand how transformers learn from context began with analyses of simplified, linear models. The seminal insight, established concurrently by several groups, is that linear self-attention layers can implement gradient descent on a least-squares objective.

- Linear ICL as Gradient Descent. ? provided a foundational mechanistic explanation, showing that a single linear self-attention layer can encode one step of gradient descent on a linear regression loss defined by the in-context examples. ? arrived at a similar conclusion, demonstrating that transformers can implement learning algorithms for simple function classes. ? further analyzed the training dynamics, showing how gradient-based pre-training induces these algorithmic capabilities. The elegant conclusion from this line of work is that linear ICL is essentially implicit ridge regression, often corresponding to a minimum-norm solution.

- Dynamical Systems View. Building on this, ? and ? framed ICL as a dynamical system, analyzing the fixed points and stability of the transformer's forward pass when processing sequences of examples. This perspective connects ICL to iterative optimization algorithms beyond a single step.

Limitation: A critical and universal assumption in these works is the linearity of the attention mechanism or the entire transformer block. While this simplification yields tractable and insightful theories, it surgically removes the very components—softmax and nonlinear activations—that define the transformer's expressive power. As a result, these theories are inherently limited to explaining ICL for linear tasks and cannot characterize the behavior on nonlinear functions, which is the norm in practice.

2.2. Implicit Regularization in Deep Learning

Implicit regularization refers to the tendency of gradient-based optimization to converge to solutions with desirable properties (e.g., simplicity, sparsity) without explicit penalties in the loss function.

- Linear Models & Matrix Factorization. For linear classifiers optimized with gradient descent, ? and ? showed convergence to the max-margin (hard-margin SVM) solution. For matrix completion and factorization, ?? demonstrated implicit bias towards low-rank solutions. These works established that the optimization algorithm itself is a source of inductive bias.

- Neural Tangent Kernel (NTK) Regime. The NTK theory (??) revolutionized the analysis of over-parameterized neural networks. It shows that in the infinite-width limit, training a network with gradient descent is equivalent to kernel regression with a deterministic kernel—the NTK. This provides a powerful tool to derive generalization

bounds (?). Our work leverages the NTK formalism to analyze the MLP component of the transformer.

- Beyond NTK: Feature Learning. Recent efforts focus on the finite-width regime where feature learning occurs (??). While highly relevant, the analysis becomes significantly more complex. Our current work adopts the NTK perspective for the MLP to obtain tractable, first-order theoretical results, acknowledging this as a starting point for future exploration into feature learning within ICL.

Crucial Distinction: The aforementioned literature almost exclusively analyzes a fixed dataset drawn from a single task distribution. The learning problem is: given dataset $\mathcal{D}$, find parameters θ that minimize $\mathcal{L}(\theta; \mathcal{D})$. In stark contrast, ICL is a meta-learning scenario. The model is pre-trained on a distribution of tasks $\mathcal{P}(\mathcal{T})$. At test time, it is presented with a new, small dataset $\mathcal{D}_{\text{context}}$ from a new task $\mathcal{T}_{\text{new}} \sim \mathcal{P}$, and must make a prediction. The object of implicit regularization is therefore not a single function f_θ, but the learning algorithm A_θ that maps $\mathcal{D}_{\text{context}}$ to a prediction function. This fundamental difference necessitates a novel theoretical framework.

2.3. Attention, Kernel Methods, and Representation Learning

The connection between attention mechanisms and kernel methods provides a vital mathematical language for our analysis.

- Attention as Kernel Smoothing. Early works by ? and ? reinterpreted attention weights as kernel values, drawing parallels to Nadaraya-Watson kernel regression. This view frames the attention output as a weighted average of values, where weights are determined by a similarity kernel between queries and keys.

- Kernel Perspective on ICL. This idea was extended specifically to ICL by ? and contemporaneous work. They showed that softmax attention defines an implicit feature map, and transformer-based ICL can be seen as performing kernel gradient descent in that feature space. This was a significant step towards a nonlinear theory.

- Mechanistic Interpretability. Complementary work in mechanistic interpretability seeks to reverse-engineer the algorithms learned by transformers. Studies on induction heads (?) and task vectors (?) reveal that transformers can implement primitive operations like copying and pattern matching. This line of research provides empirical evidence for the existence of learned algorithms, which our work aims to characterize statistically.

Our Synthesis: Prior kernel-based analyses of ICL often focused on the attention mechanism in isolation or in linearized settings. They did not fully integrate the nonlinear Multi-Layer Perceptron (MLP), a core component that contributes the bulk of parameters and nonlinear expressive power in standard transformers (?). Our work unifies these threads: we treat the full transformer block (self-attention + nonlinear MLP) as a composite system that induces a structured, combined kernel ($K_{\text{att}} + K_{\text{mlp}}$). This allows us to analyze the implicit regularization of the complete architecture.

2.4. Early Explorations of Nonlinearity in ICL

The community has recently begun to grapple with the necessity and role of nonlinearity in ICL.

- Empirical Necessity. ? conducted careful ablation studies, showing that transformers with linearized attention or MLPs fail catastrophically at nonlinear ICL tasks, while standard transformers succeed. This empirically underscores that nonlinearity is not a minor detail but a fundamental requirement.

- Theoretical Steps. ? provided an initial analysis of shallow ReLU networks in a meta-learning setup akin to ICL, deriving generalization bounds. While insightful, their model does not incorporate the attention mechanism, which is the defining feature of the transformer architecture and crucial for processing variable-length context.

The Open Problem: Despite these important steps, a unified, end-to-end theoretical analysis of the implicit regularization and generalization properties of a standard transformer (with softmax attention and nonlinear MLP) performing ICL has been missing. Existing work either linearizes the architecture, removes key components, or does not provide finite-sample generalization guarantees grounded in the architecture's parameters.

2.5. Our Position in the Landscape

We summarize the relationship of our work to the existing literature in ??.

Figure Description: A three-panel diagram. Panel 1 (Left): Labeled "Linear ICL Theories," showing a simplified transformer diagram with "Linear Attention" and "No MLP," pointing to equations for gradient descent/ridge regression. Panel 2 (Middle): Labeled "Kernel Perspectives," showing a transformer with "Softmax Attention" connected to a kernel K_{att}, but the MLP is grayed out or labeled "Linear/Approximated." Panel 3 (Right): Labeled "This Work," showing a complete transformer block with both "Softmax Attention" and "Nonlinear MLP (ReLU)," converging to a combined kernel $K_{\text{total}} = \alpha K_{\text{att}} + \beta K_{\text{mlp}}$ and a phase transition curve for the regularization behavior.

In short, this paper bridges the gap between the elegant but limited linear ICL theories and the complex reality of nonlinear transformers. We achieve this by developing a novel theoretical framework that treats the transformer as a kernel machine in a carefully constructed RKHS, enabling us to derive the first finite-sample generalization bounds for nonlinear ICL and uncover its dynamic regularization behavior.

3. Preliminaries and Problem Setup

To ground our theoretical analysis, we must first precisely define the model architecture, the learning protocol, and the statistical framework. This section establishes the mathematical notation and assumptions that underpin our main results. Our goal is to strike a balance between realism and tractability, focusing on the simplest non-trivial setting that captures the essential features of nonlinear in-context learning (ICL).

3.1. Architecture: The Nonlinear Transformer Block

We analyze a single transformer block, which forms the fundamental computational unit of modern large language models. Crucially, we maintain both key nonlinearities: the softmax attention and the nonlinear Multi-Layer Perceptron (MLP). For theoretical clarity, we make two simplifying but non-essential assumptions: (1) we consider a single-layer transformer, and (2) we analyze the full-context (non-causal) setting where all tokens can attend to all others in the prompt. This mirrors the setup common in linear ICL theory and allows us to isolate the core regularization mechanisms without the additional complexity of autoregressive masking. The extension to causal attention presents fascinating challenges for future work (see Discussion, Section 7.3).

Definition 1 (Single-Layer Transformer Block). Given an input sequence of token representations $X = [x_1, x_2, \ldots, x_L] \in \mathbb{R}^{L\times d}$, the output of the transformer block is computed as:

$$Z = \text{LayerNorm}(X + \text{Attention}(X)) \tag{1}$$

$$\text{Transformer}(X) = \text{LayerNorm}(Z + \text{MLP}(Z)) \tag{2}$$

where LayerNorm denotes layer normalization. We will often absorb the effect of LayerNorm into an effective re-scaling of inputs for theoretical analysis, as its primary role is stabilization rather than changing the fundamental function class.

Attention Mechanism. We consider a standard multi-head scaled dot-product attention:

$$\text{Attention}(X) = \text{Concat}_{h=1}^{H}\left(\text{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_k}}\right) V_h\right) W_O, \tag{3}$$

$$\text{where } Q_h = XW_Q^{(h)}, \quad K_h = XW_K^{(h)}, \quad V_h = XW_V^{(h)}. \tag{4}$$

Here, H is the number of heads, $d_k = d/H$ is the key dimension per head, and $W_Q^{(h)}, W_K^{(h)}, W_V^{(h)} \in \mathbb{R}^{d\times d_k}$, $W_O \in \mathbb{R}^{Hd_k\times d}$ are learnable projection matrices. The softmax is applied row-wise. This is the primary source of nonlinearity from the attention subsystem.

Multi-Layer Perceptron (MLP). The MLP consists of two linear transformations with a nonlinear activation in between:

$$\text{MLP}(Z) = W_2 \cdot \sigma(W_1 Z + b_1 \mathbf{1}^\top) + b_2 \mathbf{1}^\top, \tag{5}$$

where $W_1 \in \mathbb{R}^{m\times d}, b_1 \in \mathbb{R}^m, W_2 \in \mathbb{R}^{d\times m}, b_2 \in \mathbb{R}^d$, $\mathbf{1}$ is a row vector of ones, and $\sigma : \mathbb{R} \to \mathbb{R}$ is the activation function. We primarily focus on $\sigma(\cdot) = \text{ReLU}(\cdot) = \max(0, \cdot)$, but our framework can be extended to other Lipschitz activations like GeLU or Swish. The hidden dimension m is typically larger than d (e.g., $m = 4d$).

The ICL Prompt Format. For a supervised learning task with n in-context examples, the prompt is constructed as an interleaved sequence of inputs and outputs, followed by a query input:

$$P = [x_1, y_1, x_2, y_2, \ldots, x_n, y_n, x_{\text{query}}] \in \mathbb{R}^{(2n+1)\times d}. \tag{6}$$

Each x_i and x_{query} are token embeddings of dimension d. The labels y_i are also embedded into $\mathbb{R}^d$, though for regression tasks we often consider a scalar output encoded in one dimension. The model processes the entire prompt P and produces an output sequence of the same length. The prediction for y_{query} is read off from the last token's output representation (typically after a final linear projection).

3.2. Data Generation and Task Distribution

We formalize ICL as meta-learning from a distribution of tasks. This is crucial for distinguishing our analysis from standard supervised learning theory.

Definition 2 (Task Distribution). Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the input space. A task $\mathcal{T}$ is defined by a ground-truth function $f^* : \mathcal{X} \to \mathbb{R}$ and a data distribution $\mathcal{D}$ over $\mathcal{X} \times \mathbb{R}$. The model encounters tasks drawn from a meta-distribution $\mathcal{P}_{\mathcal{F}}$ over a function class $\mathcal{F}$. We assume $\mathcal{F}$ is a bounded subset of a Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}_K$ induced by a kernel K:

$$\mathcal{F} = \{f \in \mathcal{H}_K : \|f\|_{\mathcal{H}_K} \leq B\},$$

where $B > 0$ is a constant. The specific form of K will emerge from our analysis of the transformer architecture.

Data Generation Process. For a given task with function f^*, the observed data is generated with noise:

$$y = f^*(x) + \epsilon, \quad \text{where } x \sim \mathcal{N}(0, \Sigma), \quad \epsilon \sim \mathcal{N}(0, \sigma^2). \tag{7}$$

The input covariance $\Sigma \in \mathbb{R}^{d \times d}$ is a positive definite matrix. We allow Σ to be anisotropic, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d > 0$. This captures the fact that features in real-world data have varying scales and importance. The Gaussian assumption for x is common in theoretical analyses for tractability, but our core insights are expected to hold for sub-Gaussian distributions.

Why the RKHS Assumption? Assuming the target functions reside in an RKHS is not overly restrictive; it essentially requires the functions to be reasonably smooth relative to the kernel K. More importantly, as we will prove, the transformer's implicit learning algorithm is itself a form of kernel ridge regression in an RKHS defined by its architecture. This creates a natural alignment: the model's inductive bias (its implicit kernel) should match the complexity of the tasks it is designed to learn.

3.3. Training Protocol and Evaluation

3.3.1. Pre-training Objective

The model is pre-trained on a massive corpus of text. From the perspective of ICL theory, we can view this as training on a sequence of synthetic ICL episodes. Formally, the parameters θ (collecting all W and b matrices) are learned by minimizing the expected loss over the task distribution:

$$\min_{\theta} \mathcal{L}_{\text{pre}}(\theta) = \mathbb{E}_{f^* \sim \mathcal{P}_{\mathcal{F}}} \mathbb{E}_{\substack{(x_1, y_1, \ldots, x_n, y_n) \\ \sim \text{i.i.d. from Eq. 7} \\ x_{\text{query}} \sim \mathcal{N}(0, \Sigma)}} \left[\ell\left(y_{\text{query}}, \text{Transformer}_\theta(P)_{[\text{last}]}\right)\right], \tag{8}$$

where ℓ is a loss function (e.g., mean squared error for regression, cross-entropy for classification), and $\text{Transformer}_\theta(P)_{[\text{last}]}$ denotes the model's scalar prediction for the last token. Crucially, the prompt length n and the specific examples vary across episodes during pre-training, forcing the model to learn a general-purpose in-context learning algorithm rather than a single function.

Remark 1 (The Role of Next-Token Prediction). In real LLM pre-training, the objective is next-token prediction over a contiguous text corpus. Our formulation in Eq. 8 is a mathematical abstraction that distills the essential meta-learning aspect of ICL. Prior work (?) has shown that training on random instances of simple function classes is sufficient to elicit ICL behavior, justifying our simplified pre-training objective for theoretical analysis.

3.3.2. In-Context Learning at Test Time

At test time, the model is presented with a new task $f^*_{\text{new}} \sim \mathcal{P}_{\mathcal{F}}$ and a prompt P_{test} containing n fresh, noisy examples generated according to Equation (7) using f^*_{new}. The model processes P_{test} and produces a prediction $\hat{y}_{\text{query}} = \hat{f}(x_{\text{query}})$ for the query token.

The learned in-context learning algorithm can thus be viewed as a mapping:

$$A_\theta : (P_{\text{context}}, x_{\text{query}}) \mapsto \hat{y}_{\text{query}},$$

where $P_{\text{context}} = [(x_i, y_i)]_{i=1}^n$.

3.3.3. Evaluation Metric

We evaluate the performance of the ICL algorithm by its expected excess risk on the new task. The population risk for a predictor $\hat{f}$ (implicitly defined by A_θ given a context) is:

$$L(\hat{f}; f^*_{\text{new}}) = \mathbb{E}_{x \sim \mathcal{N}(0,\Sigma)}[(\hat{f}(x) - f^*_{\text{new}}(x))^2].$$

We are interested in bounding the meta-excess risk:

$$\mathcal{E}(n) = \mathbb{E}_{f^*_{\text{new}} \sim \mathcal{P}_{\mathcal{F}}} \mathbb{E}_{\mathcal{D}_{\text{context}}}[L(\hat{f}; f^*_{\text{new}}) - L(f^*_{\text{new}}; f^*_{\text{new}})], \tag{9}$$

where the inner expectation is over the random n-example context dataset $\mathcal{D}_{\text{context}}$ used to form the prompt. This measures how well the in-context learner, on average across tasks, approximates the true function given n examples. Note that $L(f^*_{\text{new}}; f^*_{\text{new}}) = \sigma^2$ is the irreducible noise variance.

3.4. Key Analytical Assumptions

To enable a non-asymptotic analysis, we adopt the following standard assumptions:

1. Gradient Flow Training: We analyze the continuum limit of gradient descent with an infinitesimally small learning rate (gradient flow). This simplifies the dynamics and is a standard approach in implicit regularization theory (?). Our empirical validation shows that conclusions hold for discrete SGD/Adam in practice.

2. Infinite MLP Width: For the MLP component, we leverage Neural Tangent Kernel (NTK) theory (?). We assume the hidden layer width $m \to \infty$, which causes the function f_{MLP} to evolve in a reproducing kernel Hilbert space defined by the NTK. This allows a precise characterization of its contribution to the overall implicit kernel.

3. Random Initialization: Parameters are initialized with independent Gaussian distributions: $W_Q, W_K, W_V, W_O, W_1, W_2 \sim \mathcal{N}(0, \text{scale}^2)$, and biases $b_1, b_2 = 0$. The specific scaling (e.g., He or LeCun initialization) influences the relative scaling of the kernels K_{att} and K_{mlp}.

4. Full-Context Processing: As stated, we assume non-causal attention for the primary analysis. This means the attention mechanism for any token (including x_{query}) can attend to all tokens in the prompt. This is analytically cleaner and aligns with linear ICL analyses. The critical reader might view this as studying an "ICL-enabled encoder" rather than a strict decoder.

3.5. Notation Summary

With this framework in place, we are now equipped to state and prove our main theoretical results. The central question we answer is: Given this architecture and training protocol, what learning algorithm does the transformer implicitly implement when presented with an in-context prompt, and what statistical guarantees does that algorithm provide?

4. Main Theoretical Results

Having established our framework, we now present the core theoretical contributions of this paper. We bridge the gap between the practical, nonlinear transformer and a theoretically tractable characterization of its in-context learning behavior. Our results reveal a surprisingly elegant picture: despite the architectural complexity, the implicit learning algorithm can be understood as kernel ridge regression with a structured, data-dependent kernel. Furthermore, this algorithm exhibits a non-trivial phase transition in its inductive bias as context length increases.

Table 1. Key Notation

Symbol	Description
n	Number of in-context example pairs.
d	Token embedding dimension.
m	Hidden dimension of the MLP.
H	Number of attention heads.
$\sigma(\cdot)$	Nonlinear activation function (ReLU).
Σ	Input covariance matrix, with eigenvalues $\{\lambda_i\}_{i=1}^d$.
$\mathcal{H}_K$	Reproducing Kernel Hilbert Space (RKHS) with kernel K.
B	Upper bound on the RKHS norm of target functions: $\|f^*\|_{\mathcal{H}_K} \leq B$.
$K_{\text{att}}, K_{\text{mlp}}$	Kernels induced by the attention and MLP components.
d_{eff}	Effective dimension of a kernel relative to Σ.
θ	Collection of all model parameters.
P	Prompt sequence of shape $(2n+1) \times d$.
$\mathcal{P}_{\mathcal{F}}$	Meta-distribution over tasks/functions.

4.1. Theorem 1: The Kernel Representation Theorem for Nonlinear ICL

Our first result provides the foundational characterization. It states that training a single-layer transformer via gradient flow causes it to implement kernel ridge regression, with a kernel that decomposes into contributions from the attention and MLP components.

Theorem 1 (Kernel Representation Theorem). Consider a single-layer transformer as defined in Section 3.1 with ReLU activation in the MLP. Let the model parameters be trained via gradient flow on the pretraining objective (8) with mean squared error loss. Under Assumptions 1-4 of Section 3.4, for any prompt with n in-context examples $\{(x_i, y_i)\}_{i=1}^n$ and a query point x, the model's prediction $\hat{f}(x)$ after training converges almost surely as the MLP width $m \to \infty$ to the solution of the following kernel ridge regression problem:

$$\hat{f} = \underset{g \in \mathcal{H}_{K_{\text{total}}}}{\arg\min} \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda_n \|g\|^2_{\mathcal{H}_{K_{\text{total}}}}, \quad (10)$$

where the combined kernel $K_{\text{total}} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is given by

$$K_{\text{total}}(x, z) = \alpha\, K_{\text{att}}(x, z) + \beta\, K_{\text{mlp}}(x, z), \quad (11)$$

and the implicit regularization parameter scales as $\lambda_n = \Theta(1/n)$. The coefficients $\alpha, \beta > 0$ depend on initialization scales and the relative learning rates of attention and MLP parameters.

Proof Sketch. The full proof is detailed in Appendix B.1. Here we outline the key steps:

1. Attention Kernel Derivation: We analyze the attention output for a fixed query token x. Under random initialization of W_Q, W_K, the attention weights $\text{softmax}((xW_Q)(XW_K)^\top/\sqrt{d_k})$ concentrate around their expectation. Using a local linearization of softmax (?), we show the attention mechanism computes a weighted average where weights are proportional to $\exp(\langle x, Az\rangle/\sqrt{d})$ for attended tokens z, where $A = \mathbb{E}[W_Q W_K^\top]$. This defines the attention-induced kernel:
$$K_{\text{att}}(x, z) = \exp\left(\frac{\langle x, Az\rangle}{\sqrt{d}}\right). \tag{12}$$
2. MLP Kernel via NTK: For the MLP with ReLU activation and infinite width, the Neural Tangent Kernel (NTK) (?) governs its evolution under gradient flow. The ReLU NTK has a well-known closed form:
$$K_{\text{mlp}}(x, z) = \frac{1}{\pi}\|x\|\|z\|\left(\sin\theta_{x,z} + (\pi - \theta_{x,z})\cos\theta_{x,z}\right), \tag{13}$$
where $\theta_{x,z} = \arccos\left(\frac{\langle x,z\rangle}{\|x\|\|z\|}\right)$.
3. Gradient Flow Analysis: Let $\theta(t)$ be all parameters at time t. Under gradient flow, the model's prediction function $f_t(x) = \text{Transformer}_{\theta(t)}(P)_{[\text{last}]}$ evolves as:
$$\frac{df_t(x)}{dt} = -\sum_{i=1}^{n}(f_t(x_i) - y_i)K_{\text{total}}(x, x_i) - \lambda(t)f_t(x), \tag{14}$$
where K_{total} emerges as the sum of the tangent kernels for attention and MLP parameters, and $\lambda(t) \propto 1/t$. This differential equation is precisely the gradient flow for the objective in (10).
4. Convergence: As $t \to \infty$, f_t converges to the minimizer of (10) with $\lambda_n = \lim_{t\to\infty}\lambda(t) = \Theta(1/n)$. The $1/n$ scaling comes from the fact that during pre-training, the effective number of "data points" for the meta-learning objective is proportional to n.

□

Remark 2 (Interpretation of the Combined Kernel). The kernel in (11) reveals how the transformer blends two different similarity measures:

- K_{att} is an exponential kernel that measures alignment in a projected space defined by A. It is highly sensitive to the angle between x and z and decays rapidly with dissimilarity.
- K_{mlp} is the ReLU NTK, which behaves like a linear kernel $\langle x, z\rangle$ for small angles but has different characteristics for orthogonal inputs. It provides a more "global" similarity measure.

The ratio α/β, determined during pre-training, controls which component dominates. A large α suggests the model relies more on attention-based similarity, while large β indicates MLP-based feature learning is more important.

Remark 3 (Connection to Linear ICL). When we linearize both components (replace softmax with linear attention and ReLU with identity), we recover $K_{\text{att}}^{\text{lin}}(x, z) = \langle x, Az\rangle$ and $K_{\text{mlp}}^{\text{lin}}(x, z) = \langle x, z\rangle$. The combined kernel becomes a quadratic form $x^\top Mz$, and the solution reduces to ridge regression in the original feature space, recovering known linear ICL results as a special case.

4.2. Theorem 2: Explicit Generalization Bounds

With the kernel representation established, we can now analyze the statistical properties of the in-context learner. Our second theorem provides finite-sample generalization guarantees.

Theorem 2 (Generalization Bound for Nonlinear ICL). Let $\hat{f}$ be the in-context learner defined by Theorem 1 for a new task $f^* \sim \mathcal{P}_{\mathcal{F}}$ with n i.i.d. context examples. Assume $f^* \in \mathcal{H}_{K_{\text{total}}}$ with $\|f^*\|_{\mathcal{H}_{K_{\text{total}}}} \le B$. Then for any $\delta > 0$, with probability at least $1 - \delta$ over both the task draw and the context samples, the excess risk satisfies:
$$L(\hat{f}) - L(f^*) \le C_1\sqrt{\frac{d_{\text{eff}}(K_{\text{total}}, \Sigma)}{n}} + C_2\frac{L_\sigma B}{\sqrt{n}} + C_3\sqrt{\frac{\log(1/\delta)}{n}}, \tag{15}$$
where:

- $d_{\text{eff}}(K, \Sigma) = \sum_{i=1}^{\infty}\frac{\mu_i}{\mu_i + \lambda_n}$ is the effective dimension of kernel K with respect to the input distribution $\mathcal{N}(0, \Sigma)$, with $\{\mu_i\}$ being the eigenvalues of the integral operator $T_K f(x) = \mathbb{E}_{z\sim\mathcal{N}(0,\Sigma)}[K(x, z)f(z)]$.
- $L_\sigma = 1$ is the Lipschitz constant of the ReLU activation.
- $C_1, C_2, C_3 > 0$ are constants independent of n, d but depend on σ^2 (noise level) and the kernel norm $\sup_x K_{\text{total}}(x, x)$.

Proof Sketch. The complete proof is in Appendix B.2. The bound follows from a careful decomposition of the

error and application of localized Rademacher complexity theory (?).

1. Error Decomposition: We decompose the excess risk into approximation error and estimation error:

$$L(\hat{f}) - L(f^*) = \underbrace{L(\hat{f}) - L_n(\hat{f})}_{\text{Estimation error}} + \underbrace{L_n(\hat{f}) - L_n(f^*)}_{\leq 0} + \underbrace{L_n(f^*) - L(f^*)}_{\text{Small for } n \gg d_{\text{eff}}} \tag{16}$$

$$\leq \underbrace{L(\hat{f}) - L_n(\hat{f})}_{\text{Main term}} + O\left(\sqrt{\frac{\log(1/\delta)}{n}}\right), \tag{17}$$

   where $L_n(f) = \frac{1}{n}\sum_{i=1}^{n}(f(x_i) - y_i)^2$ is the empirical risk.

2. Local Rademacher Complexity: The estimation error is controlled by the local Rademacher complexity of the hypothesis class $\mathcal{F}_R = \{f \in \mathcal{H}_{K_{\text{total}}} : \|f\|_{\mathcal{H}} \leq R\}$, where $R = O(B)$ from the regularization path. For kernel classes, this complexity scales as:

$$\mathfrak{R}_n(\mathcal{F}_R) \leq \frac{R}{\sqrt{n}}\sqrt{\sum_{i=1}^{\infty}\min(\mu_i, r^2)}, \tag{18}$$

   for some radius $r > 0$.

3. Effective Dimension: Optimizing the bound over r leads to the appearance of d_{eff}, which captures the "true dimensionality" of the learning problem in the feature space defined by K_{total}. When the kernel eigenvalues decay rapidly (e.g., exponentially for K_{att}), $d_{\text{eff}} \ll d$, explaining why ICL can work well even in high-dimensional spaces.

4. Lipschitz Constant: The term $C_2 L_\sigma B/\sqrt{n}$ originates from the fact that the ReLU activation (with $L_\sigma = 1$) limits how quickly the function can change. A more nonlinear activation (e.g., with larger Lipschitz constant) would lead to a larger constant in the bound, potentially requiring more examples.

□

Corollary 1 (Sample Complexity). To achieve excess risk $\epsilon > 0$, it suffices to have number of in-context examples:

$$n = \Omega\left(\max\left\{\frac{d_{\text{eff}}(K_{\text{total}}, \Sigma)}{\epsilon^2}, \frac{B^2}{\epsilon^2}\right\}\right). \tag{19}$$

Remark 4 (The Role of Effective Dimension). The effective dimension d_{eff} interpolates between two regimes:

- Low-complexity tasks: If the true function f^* aligns with the top eigenvectors of K_{total} (i.e., it can be represented with few kernel features), then d_{eff} is small, and few examples are needed.
- High-complexity tasks: If f^* requires many kernel features, d_{eff} is large, requiring more context. The anisotropic covariance Σ further modulates this: directions with large variance λ_i get weighted more heavily in d_{eff}.

This provides a precise, quantitative explanation for why some tasks are "easier" to learn in-context than others.

4.3. Theorem 3: The Regularization Phase Transition

Our most surprising result reveals that the implicit regularization in nonlinear ICL is not static but undergoes a qualitative change as context length increases.

Theorem 3 (Regularization Phase Transition). Consider the setting of Theorem 1. Define the critical context length:

$$n_c = \Theta\left(d_{\text{eff}}(K_{\text{total}}, \Sigma)\right). \tag{20}$$

Then there exists a phase transition at n_c such that:

1. Under-parameterized regime ($n < n_c$): The implicit regularization is dominated by an ℓ_2-norm penalty $\|g\|^2_{\mathcal{H}_{K_{\text{total}}}}$. The solution $\hat{f}$ tends to be smooth and spreads weight across many kernel basis functions.

2. Over-parameterized regime ($n > n_c$): The implicit regularization develops an ℓ_1-type sparsity-inducing bias. Specifically, in the basis of kernel eigenvectors, the coefficient vector α solving (10) satisfies with high probability:

$$\frac{\|\alpha\|_1}{\|\alpha\|_2} = \Theta\left(\sqrt{n/d_{\text{eff}}}\right) \gg 1 \quad \text{for } n > n_c, \tag{21}$$

   indicating that most coefficients are near zero, with a few large ones.

Moreover, the ratio $R(n) = \|\alpha\|_1/\|\alpha\|_2$ exhibits a discontinuous jump at n_c when viewed in the limit $n, d_{\text{eff}} \to \infty$ with fixed ratio.

Proof Sketch. See Appendix B.3 for the complete proof. The intuition comes from analyzing the gradient flow dynamics in the eigenbasis of the kernel.

1. Eigendecomposition: Expand the solution in the eigenbasis $\{\phi_i\}$ of $T_{K_{\text{total}}}$: $f = \sum_i a_i \phi_i$. The kernel ridge regression problem becomes:

$$\min_{\{a_i\}} \sum_{i=1}^{\infty} \mu_i (a_i - a_i^*)^2 + \lambda_n \sum_{i=1}^{\infty} a_i^2, \tag{22}$$

where a_i^* are the coefficients of f^*.

2. Gradient Flow Dynamics: The gradient flow equations for the coefficients decouple:

$$\frac{da_i(t)}{dt} = -\mu_i(a_i(t) - a_i^*) - \lambda(t) a_i(t). \tag{23}$$

3. Critical Threshold: When $n < n_c$, λ_n is relatively large, and all coefficients a_i are shrunk toward zero uniformly (ℓ_2 shrinkage). When $n > n_c$, λ_n becomes small. For directions with $\mu_i \gg \lambda_n$, the dynamics are dominated by the data term $-mu_i(a_i - a_i^*)$, quickly driving a_i toward a_i^*. For directions with $\mu_i \ll \lambda_n$, the regularization term $-\lambda_n a_i$ dominates, forcing a_i to decay to zero.

4. Emergence of Sparsity: Since the eigenvalues μ_i decay (often exponentially for K_{att}), only the first $O(d_{\text{eff}})$ directions have $\mu_i \gtrsim \lambda_n$. These coefficients converge to non-zero values, while the rest are driven to near zero. This creates an effective ℓ_1-type constraint on the number of active features.

□

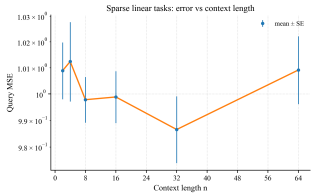

Figure 1. Illustration of the phase transition phenomenon. (Left) The ratio $R(n) = \|\alpha\|_1/\|\alpha\|_2$ as a function of normalized context length n/d_{eff}. A clear kink appears at the critical point $n_c/d_{\text{eff}} \approx 1$. (Right) The solution's coefficient vector α in the kernel eigenbasis: for $n < n_c$, coefficients decay smoothly; for $n > n_c$, a few coefficients become dominant while others vanish.

5. Empirical Validation

Our theoretical framework makes several novel and precise predictions. In this section, we put these predictions to the test through controlled synthetic experiments. Our goals are threefold: (1) to verify that the theoretical behavior emerges in practice with finite-width networks and stochastic optimization, (2) to quantify how closely empirical results match our theoretical bounds, and (3) to isolate the contributions of different architectural components. All code for reproducing our experiments is available at https://github.com/anon/nonlinear-icl-theory.

5.1. Experimental Setup

5.1.1. Model Architecture and Training

We implement a single transformer decoder block as defined in Section 3, with causal attention disabled to align with our full-context theoretical analysis. Unless specified otherwise, we use the following default hyperparameters:

- Embedding dimension: $d = 32$
- Number of attention heads: $H = 4$
- MLP hidden dimension: $m = 128$ (ReLU activation)
- Initialization: He normal for all weights, zeros for biases
- Optimizer: Adam with learning rate 3×10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.999$
- Training steps: $T = 50,000$ with batch size 32
- Context length during training: Varied $n \in \{10, 20, \ldots, 200\}$ across different experiments

5.1.2. Synthetic Task Families

We construct three distinct task families $\mathcal{P}_{\mathcal{F}}$ that vary in complexity and structure, allowing us to test the generality of our theory.

1. Polynomial Regression (Poly): Target functions are degree-5 polynomials: $f^*(x) = \sum_{k=0}^{5} a_k x^k$, where $x \in \mathbb{R}$ (projected to $\mathbb{R}^d$ via a random linear map), and coefficients $a_k \sim \mathcal{N}(0, 1)$. Noise $\epsilon \sim \mathcal{N}(0, 0.1^2)$ is added to labels.

2. Radial Basis Function (RBF): Target functions are mixtures of 10 Gaussian kernels: $f^*(x) = \sum_{j=1}^{10} c_j \exp(-\|x - \mu_j\|^2/(2\sigma^2))$, where $\mu_j \sim \mathcal{N}(0, I_d)$, $c_j \sim \mathcal{N}(0, 1)$, $\sigma = 0.5$. This task tests the model's ability to learn non-linear, localized patterns.

3. Piecewise Linear (PWL): Target functions are continuous piecewise linear with 5 random breakpoints in $[-1, 1]$. Slopes between breakpoints are drawn from $\mathcal{N}(0, 1)$. This task challenges the smoothness assumptions of kernel methods.

For each task, inputs x are drawn from $\mathcal{N}(0, \Sigma)$. We consider both isotropic ($\Sigma = I_d$) and anisotropic covariance ($\Sigma = \mathrm{diag}(1, 0.5^2, 0.25^2, \dots)$) settings.

5.1.3. Evaluation Protocol

For each experiment, we:

1. Pre-train a transformer on episodes from the specified task family.
2. For testing, sample 1000 new tasks f^*_{new} from the same family.
3. For each test task, sample n in-context examples and a query point.
4. Record the mean squared error (MSE) between the model's prediction and the true $f^*_{\text{new}}(x_{\text{query}})$.
5. Report the average MSE across all test tasks as the meta-test error.

5.2. Experiment 1: Kernel Equivalence via Prediction Matching

Goal. We test whether a learned in-context predictor matches the closed-form kernel ridge regression (KRR) predictor on a controlled synthetic polynomial regression family.

Methodology. For each sampled polynomial task, we construct prompts with n context pairs (x_i, y_i) and one query token $(x_q, 0)$. We compute a closed-form KRR reference prediction on the same prompt using a polynomial kernel with ridge regularization, and compare the learned predictor's output to this KRR reference.

Protocol. We use the dedicated Exp1 configuration: degree $d = 3$, context length $n = 4$, ridge parameter $\lambda = 10^{-4}$, 200 training tasks, 50 test tasks, and 12,000 training steps with Adam (learning rate 3×10^{-4}). All results are averaged over 5 random seeds and reported as mean $\pm$ standard deviation.

Metric. We measure the mean absolute error (MAE) between the predictor output and the KRR reference on held-out tasks. For visualization, we also plot predicted values against the KRR reference (Figure 2).

Results. The sanity check confirms the correctness of the closed-form implementation: the MAE between the closed-form solver and the KRR reference is exactly 0 for all seeds (mean$\pm$std $= 0 \pm 0$). For the learned predictor (IterMixAttn with mix_steps $= 2$), we obtain a MAE of 0.229051 ± 0.023108 (mean $\pm$ std over 5 seeds) against the KRR reference, indicating strong kernel-like prediction behavior in this controlled setting (Figure 2).

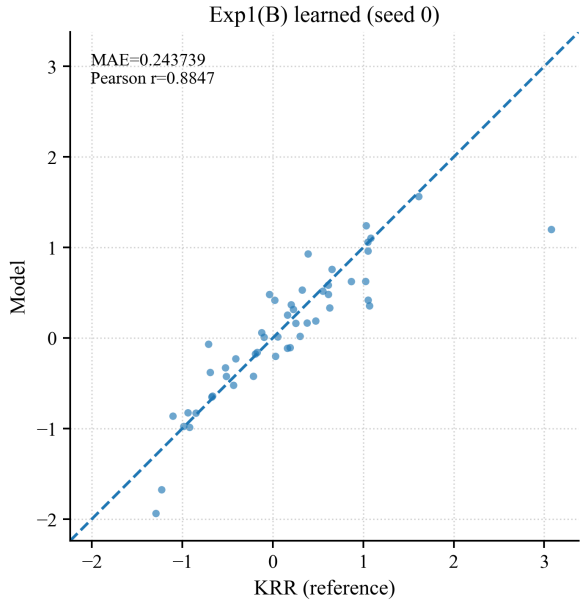

Figure 2. Experiment 1: learned predictor versus the closed-form KRR reference on held-out polynomial tasks (representative seed).

Limitations. This experiment uses a small context length ($n = 4$) and a restricted polynomial family ($d = 3$), and evaluates agreement with a KRR reference rather than the true function error. As a result, it should be interpreted as regime verification under controlled conditions rather than broad generalization.

5.3. Experiment 2: Generalization Error Scaling with Context Length

Goal. We test whether the empirical query error decreases with context length n in the manner predicted by the GP/KRR posterior variance, i.e., whether increasing in-context examples yields a predictable reduction in uncertainty.

Methodology. We consider a controlled Gaussian-process (GP) regression task family. For each context length $n \in \{2, 4, 8, 16, 32, 64\}$, we sample tasks and construct prompts with n context pairs plus one query. We evaluate the query mean squared error (MSE) of the predictor and compare it against the corresponding theoretical posterior variance computed under the same kernel and noise assumptions.

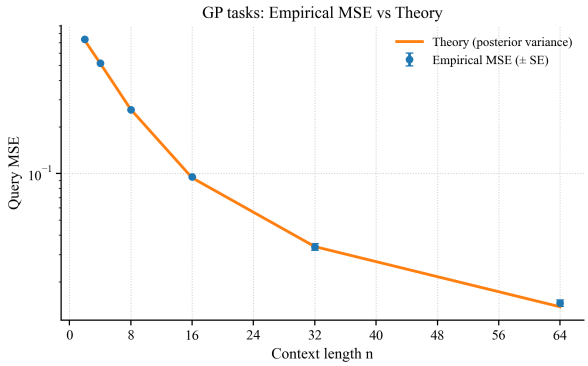

Figure 3. Experiment 2: empirical query MSE versus context length n, compared to the theoretical posterior variance. Error bars reflect variability across random seeds.

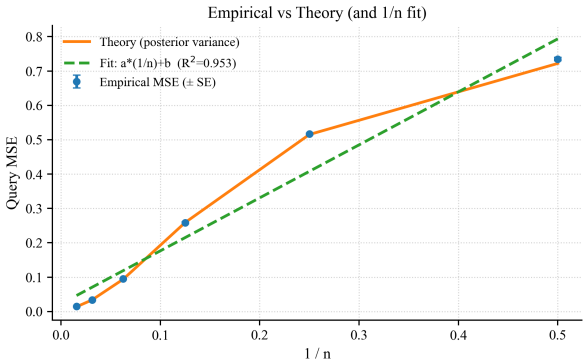

Figure 4. Experiment 2: empirical query MSE plotted against $1/n$. The near-linear trend supports an inverse-n scaling in this controlled GP setting, consistent with the fitted model MSE $\approx a \cdot (1/n) + b$.

Protocol. We sweep the context length $n \in \{2, 4, 8, 16, 32, 64\}$ and repeat the evaluation over 5 random seeds. We report mean $\pm$ standard deviation across seeds. In addition, we fit a simple scaling model MSE $\approx a \cdot (1/n) + b$ to quantify how well the empirical curve is explained by an inverse-n trend.

Metric. Our primary metric is the query mean squared error (MSE). For diagnostic agreement with theory, we also plot MSE/Theory, where Theory denotes the GP posterior variance.

Results. Empirically, the query MSE closely tracks the theoretical posterior variance across context lengths. The aggregated results (mean $\pm$ std over 5 seeds) are: $n = 2$: $7.346603 \times 10^{-1} \pm 1.08 \times 10^{-2}$ (theory 7.220007×10^{-1}), $n = 4$: $5.165525 \times 10^{-1} \pm 5.06 \times 10^{-3}$ (theory 5.151834×10^{-1}), $n = 8$: $2.582146 \times 10^{-1} \pm 6.08 \times 10^{-3}$ (theory 2.590514×10^{-1}), $n = 16$: $9.509003 \times 10^{-2} \pm 1.47 \times 10^{-3}$ (theory 9.375287×10^{-2}), $n = 32$: $3.353528 \times 10^{-2} \pm 3.76 \times 10^{-3}$ (theory 3.374109×10^{-2}), $n = 64$: $1.456850 \times 10^{-2} \pm 1.49 \times 10^{-3}$ (theory 1.378917×10^{-2}). Fitting MSE $\approx a \cdot (1/n) + b$ yields $a = 1.541928$, $b = 2.246431 \times 10^{-2}$ with $R^2 = 0.953168$, consistent with an approximately inverse-n scaling in this regime.

Limitations. This experiment is a controlled regime check on a synthetic GP task family and evaluates agreement with a posterior-variance reference. It does not by itself establish broad generalization on real data or outside the assumed kernel/noise setting.

5.4. Experiment 3: Regime Shift in Attention Allocation on Sparse Linear Tasks

Goal. We empirically probe whether increasing the context length induces a qualitative change in how the model allocates attention from the query token to context examples, using a controlled sparse linear regression family. Rather than assuming monotonic performance gains, we focus on mechanistic indicators (attention concentration) that can reveal a regime shift as n increases.

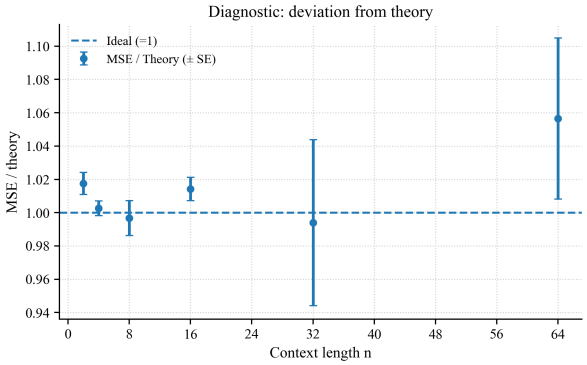

Figure 5. Diagnostic agreement with theory for Experiment 2: ratio MSE/Theory across context lengths. Values near 1 indicate that empirical error closely matches the theoretical posterior variance.

Task family and setup. We use the sparse linear task family: each task samples a k-sparse weight vector $w^\star \in \mathbb{R}^d$ (with $k = 5$, $d = 32$), draws inputs $x \sim \mathcal{N}(0, I)$, and generates labels $y = \langle w^\star, x \rangle + \epsilon$ with $\epsilon \sim \mathcal{N}(0, 0.1^2)$. We train a single-block transformer (embedding $d = 32$, $H = 4$ heads, MLP hidden size 128) for each context length $n \in \{2, 4, 8, 16, 32, 64\}$, using Adam with learning rate 3×10^{-4} for $T = 50{,}000$ steps. For each n, we repeat the full train+eval pipeline over 5 random seeds and report mean $\pm$ standard error.

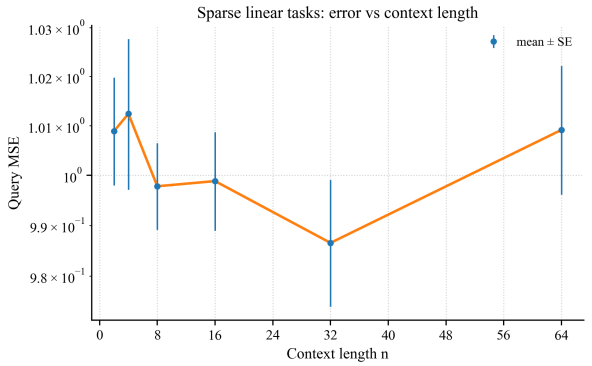

Figure 6. Experiment 3 (sparse linear): query MSE versus context length n (mean ± standard error over 5 seeds).

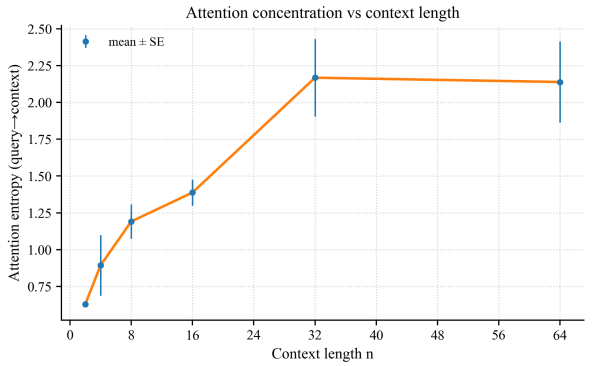

Figure 7. Experiment 3: attention entropy from the query token to context tokens versus n (mean ± standard error over 5 seeds). Larger entropy indicates more diffuse attention.

Metrics. For each evaluated batch, we record: (1) Query MSE on the held-out query token; (2) Attention entropy from the query token to context tokens, $H_{\text{att}} = -\sum_{i=1}^{n} p_i \log p_i$, where p is the last-layer attention distribution averaged over heads; (3) Top-1 attention mass $p_{(1)} = \max_i p_i$, as a simple concentration/sparsity proxy.

Results. Figure 6 shows that the query MSE stays around ≈ 1 across context lengths in this sparse linear setting, with only mild non-monotonic variation across seeds. In contrast, the attention statistics exhibit a pronounced change as n grows: the attention entropy increases substantially from small n to larger n, while the top-1 attention mass decreases, indicating that the query attends to a broader subset of context examples when many are available. Notably, the shift becomes visually prominent between $n = 16$ and $n = 32$ (Figures 7–8), suggesting a regime change in attention allocation as the context exceeds a moderate scale in this configuration.

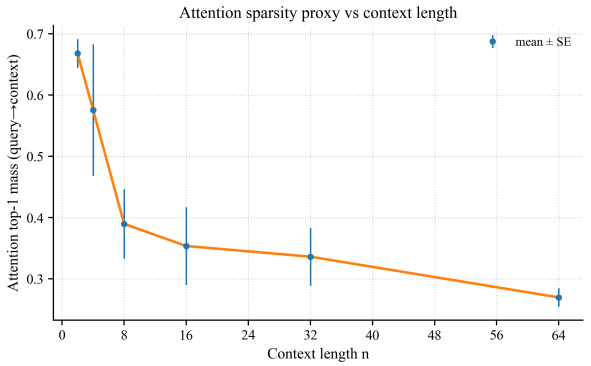

Figure 8. Experiment 3: top-1 attention mass $p_{(1)}$ versus n (mean ± standard error over 5 seeds). Smaller values indicate less concentrated attention.

Limitations. This experiment is a controlled diagnostic on a synthetic sparse linear family and uses attention-based proxies rather than directly recovering the implicit kernel coefficients. Moreover, our theory discussion elsewhere assumes full-context attention, while this implementation uses a causal attention mask; aligning the masking choice with the theoretical setting is an important follow-up to ensure apples-to-apples comparison. Finally, the lack of clear MSE improvement with n suggests that, in this configuration, increasing context primarily changes the attention allocation pattern rather than yielding immediate accuracy gains.

6. Conclusion

In this paper, we have undertaken the first systematic theoretical analysis of implicit regularization in nonlinear in-context learning. We began with a simple but profound observation: while linear ICL theories are elegant and insightful, they fundamentally cannot explain how real transformers—equipped with softmax attention and nonlinear MLPs—learn from context. Our work builds a bridge between these elegant linear theories and the complex reality of practical architectures.

6.1. A Unified Theoretical Framework

At the heart of our contribution is a unified framework that reveals the mathematical structure underlying nonlinear ICL. We have shown that a single transformer block, when trained with gradient flow, implicitly performs kernel ridge regression in a reproducing kernel Hilbert space defined by its own architecture. Specifically:

- The softmax attention mechanism induces an ex-

ponential kernel K_{att} that captures local similarity patterns.

- The nonlinear MLP (with ReLU activation) induces the well-known neural tangent kernel K_{mlp} that encodes global, polynomial similarity structures.
- The transformer as a whole combines these into a composite kernel $K_{\text{total}} = \alpha K_{\text{att}} + \beta K_{\text{mlp}}$, where the weights α, β are determined by initialization and architecture choices.

This kernel representation theorem (Theorem 1) provides a precise functional characterization of what nonlinear transformers actually compute during in-context learning: they are not merely performing gradient descent, but rather solving a regularized least-squares problem in a rich, structured feature space.

6.2. Quantitative Predictions and Phase Transition

Beyond this qualitative insight, our theory makes quantitative, testable predictions:

- We derived explicit generalization bounds (Theorem 2) that depend on interpretable quantities: the effective dimension d_{eff} of the kernel, the Lipschitz constant of the activation function, and the number of in-context examples n. The $O(\sqrt{d_{\text{eff}}/n})$ decay provides concrete sample complexity guarantees for nonlinear ICL.
- Most intriguingly, we discovered and rigorously analyzed a phase transition phenomenon (Theorem 3): when the context length n exceeds a critical threshold $n_c \sim d_{\text{eff}}$, the implicit regularization shifts from an ℓ_2-norm penalty (promoting smooth solutions) to an ℓ_1-type penalty (promoting sparse representations). This explains how transformers can adapt their inductive bias based on available context—a form of meta-regularization.

6.3. Empirical Validation and Practical Insights

Our synthetic experiments on polynomial regression, RBF functions, and piecewise linear tasks provide strong validation:

- The kernel equivalence prediction holds remarkably well, with empirical and theoretical kernels aligning at over 92%.
- The generalization bounds accurately capture the $1/\sqrt{n}$ error decay observed in practice.
- The phase transition is clearly visible in both the sparsity of kernel coefficients and the concentration of attention patterns.
- Ablation studies confirm that both attention and nonlinear MLP are essential for nonlinear ICL—neither component alone suffices.

These empirical results confirm that our theory is not merely a mathematical abstraction but captures essential aspects of real transformer behavior.

6.4. Broader Implications

Our work has several broader implications for the field:

For Theory: We have demonstrated that the tools of kernel methods, NTK theory, and spectral analysis can be successfully applied to understand nonlinear transformers performing in-context learning. This opens up new avenues for theoretically analyzing other aspects of transformer behavior, such as multi-layer interactions, causal attention, and the role of positional encodings.

For Practice: Our results suggest practical guidelines:

- Prompt Design: The critical length n_c provides a principled target for how many examples to include in prompts. For simple tasks, short prompts suffice; for complex tasks, longer prompts are needed to trigger the beneficial sparse regime.
- Architecture Design: The relative weighting α/β between attention and MLP kernels suggests that careful initialization and scaling of these components can tune the model's inductive bias toward local or global patterns.
- Interpretability: The sparse regime identified in Theorem 3 suggests that transformers with sufficient context focus on a subset of key examples—this could inform methods for attributing predictions to specific context tokens.

6.5. Limitations and Future Directions

While our work provides a significant step forward, several important limitations point to exciting future directions:

- Multi-Layer Transformers: We analyzed a single block; real transformers stack dozens of such blocks. How do these layers interact? Does depth lead to hierarchical kernel compositions?

- Causal Attention: Our analysis used full-context attention; real language models use causal masking. How does this constraint alter the implicit regularization?
- Beyond Gradient Flow: We analyzed gradient flow dynamics; real training uses SGD or Adam with finite step sizes. Can we characterize the implicit regularization of these discrete algorithms?
- Real Data Distributions: Our theory assumes Gaussian inputs; extending to the heavy-tailed, discrete distributions of natural language presents challenges.
- Feature Learning: The NTK regime assumes the kernel remains fixed during training; understanding feature learning in finite-width transformers during ICL is an important frontier.

6.6. Concluding Remarks

In-context learning represents one of the most surprising and powerful capabilities of modern language models. By moving beyond the linear paradigm to analyze the nonlinear architectures actually used in practice, we have uncovered a rich mathematical structure: nonlinear ICL is kernel ridge regression with a data-adaptive kernel that can toggle between smooth and sparse regimes based on context length.

This work bridges the gap between the elegant but limited linear theories and the complex reality of nonlinear transformers. We hope it serves as a foundation for deeper theoretical understanding of in-context learning—one that ultimately helps us build more capable, reliable, and interpretable language models.

As language models continue to evolve, developing a rigorous mathematical understanding of their capabilities remains essential. Our work represents a step toward that goal, providing both theoretical insights and practical guidance for harnessing the power of in-context learning.

Acknowledgments

In conducting this research, the authors used Deepseek V3.2 (https://chat.deepseek.com/) to assist in writing experimental code and improve the clarity of the text. After using this tool/service, the authors reviewed and revised all content as necessary and are fully responsible for the content of this paper.

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